

RESEARCH PROPOSAL

Spectroscopic Dissection of Type Ia Supernovae for Precision Cosmological Distance Measurements

*Toward Resolving the H_0 Tension and Constraining Dark Energy Evolution,
grounded in first-principles, quantum-accelerated explosion physics*

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Executive Summary. Type Ia supernovae (SNe Ia) are the cornerstone of cosmological distance measurement, yet the empirical *Phillips relation* used to standardize their luminosities may conceal physically distinct subclasses with systematically different intrinsic brightnesses. Recent observations of “optical twins”—SN Ia pairs with nearly identical optical spectra but discrepant distance moduli—support this hypothesis, with their divergence traced to differences in near-ultraviolet (NUV) luminosity. This proposal will (1) develop a GPU-accelerated spectral-analysis framework (**Lumina-SN/Lumina-ML**) to extract the physical parameters of SNe Ia from their spectra; (2) identify and characterize NUV-divergent subclasses with the upcoming **S-Drift** space-UV survey; (3) construct a refined, subclass-aware absolute-luminosity calibration that improves H_0 and dark-energy constraints; and (4) **ground the subclass physics in a first-principles, quantum-accelerated explosion-physics module**, in which **IonQ** (gate-model) and **D-Wave** (annealer) quantum processors attack the two computationally hardest pieces of the deflagration-to-detonation transition (DDT)—the nuclear reaction microphysics and the combinatorial ignition / inverse problems—that ultimately set the ^{56}Ni mass and iron-group distribution responsible for the observed subclasses.

1 Scientific Background and Motivation

1.1 Type Ia Supernovae as Standardizable Candles

Type Ia supernovae arise from the thermonuclear disruption of carbon–oxygen white dwarfs in binary systems. Their utility in cosmology stems from the Phillips relation (Phillips 1993): the peak luminosity of an SN Ia correlates with the rate at which its light curve declines, parameterized by $\Delta m_{15}(B)$, the B -band magnitude drop in the first 15 days after maximum. This single-parameter correction transforms SNe Ia into standardizable candles, enabling precise distances across cosmological scales.

The application of this relation led to the landmark discovery of cosmic acceleration (Riess et al. 1998; Perlmutter et al. 1999), establishing dark energy as a central problem in modern physics. Current SN Ia distances reach $\sim 5\text{--}7\%$ statistical precision per object after standardization. However, as Rubin/LSST and Roman aim to constrain the dark-energy equation of state to percent-level precision, the *systematic* floor of the Phillips relation itself becomes the limiting factor.

Key Question

Is the Phillips relation a single, universal relation, or does it conceal physically distinct subclasses of SNe Ia with systematically different luminosity calibrations?

1.2 The Subclass Hypothesis: Cracks in the Phillips Relation

Mounting evidence suggests the Phillips relation may not describe a single physical sequence:

- **Progenitor-channel diversity:** single-degenerate (accretion) and double-degenerate (WD merger) channels may yield subtly different nucleosynthetic yields and spectral properties even at fixed light-curve shape.
- **Residual Hubble scatter:** after Phillips and color corrections, the residual scatter ($\sigma \approx 0.10\text{--}0.15$ mag) exceeds photometric errors, implying intrinsic diversity uncaptured by current parameters.
- **Spectroscopic diversity:** the Wang et al. (2009) split into “Normal,” “High-Velocity” (HV), and 91T/91bg-like groups (via Si II $\lambda 6355$ velocity) shows distinct populations within the “normal” class; HV events prefer massive hosts and may have different Hubble residuals.

1.3 Optical Twins and NUV Divergence: A Smoking Gun

Observational Evidence

“Optical twins”—SN Ia pairs with nearly identical optical spectra (3500–7500 Å) but measurably different distance moduli—have been identified. Crucially, these twins diverge in their *near-ultraviolet* (NUV; 2000–3500 Å) luminosity, indicating that the NUV carries independent information about the explosion physics not encoded in the optical.

The striking example is SN 2011fe and SN 2011by (Foley & Kirshner 2013; Graham et al. 2015): near-identical optical spectra, Δm_{15} , and colors, yet SN 2011by was ~ 0.335 mag fainter at peak in all optical bands. HST UV spectroscopy revealed dramatic divergence below ~ 3500 Å, with SN 2011by showing suppressed NUV flux (Figure 1). Foley & Kirshner attributed this to a factor ~ 1.7 difference in ^{56}Ni mass, though Graham et al. found near-identical nebular spectra, suggesting progenitor metallicity or iron-group distribution rather than total ^{56}Ni mass alone.

Milne et al. (2013, 2015) used Swift/UVOT to show normal SNe Ia separate into NUV-red and NUV-blue subgroups (the latter $\sim 1/3$ of the sample), offset by ~ 0.4 mag in NUV–optical color; the NUV-blue group shows less iron-peak absorption and unburned carbon. Burns et al. (2020) studied “sibling” SNe in NGC 5643 (an ideal controlled comparison), and Riess et al. (2024) demonstrated spectroscopic twins toward a precise H_0 .

The NUV ($\lambda \lesssim 3500$ Å) is dominated by a dense forest of iron-group element (IGE) lines—Fe II, Fe III, Co II, Ni II—highly sensitive to: (1) the total ^{56}Ni mass (which sets the peak luminosity via $^{56}\text{Ni} \rightarrow ^{56}\text{Co} \rightarrow ^{56}\text{Fe}$);

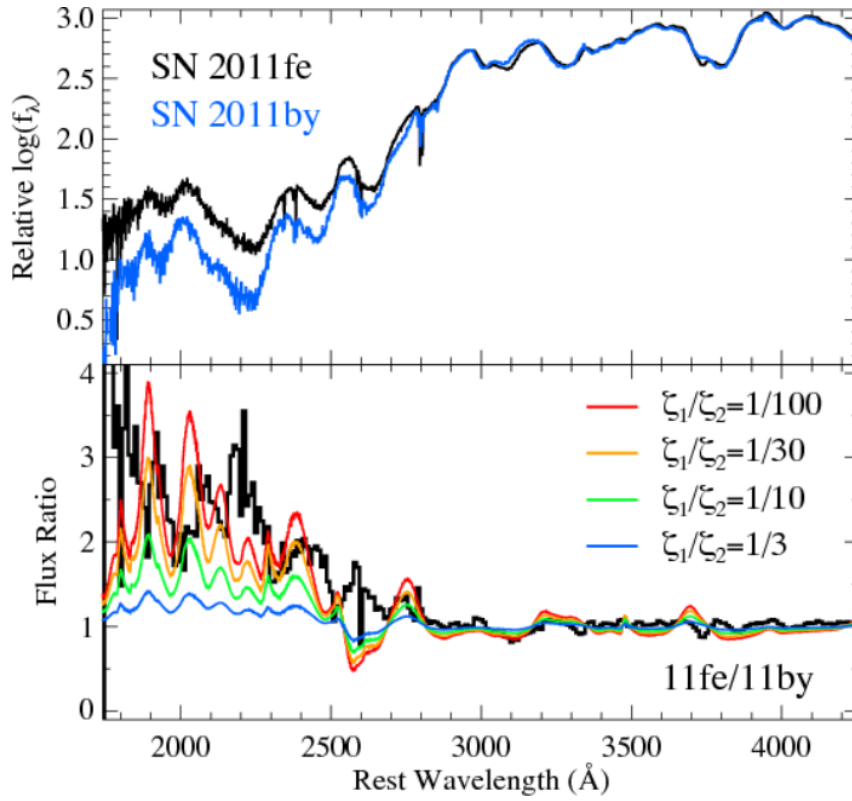


Figure 1: UV-optical spectra of the optical twins SN 2011fe (black) and SN 2011by (blue). *Top*: near-identical at optical wavelengths ($\lambda > 3500 \text{ \AA}$) but dramatically divergent in the NUV. *Bottom*: flux ratio (11fe/11by); colored curves show progenitor-metallicity predictions ζ_1/ζ_2 (Lentz et al. 2000). The NUV encodes the broken degeneracy. (After Graham et al. 2015, MNRAS 446, 2073.)

(2) the *radial distribution* of IGE in the ejecta (which sets the opacity structure and the emergent UV flux); and (3) the progenitor metallicity (which fixes the initial abundance of stable iron-peak species, e.g. ^{54}Fe , ^{58}Ni , that blanket the UV without changing the radioactive budget). Two SNe Ia can thus share an optical spectrum (set by Si, S in the outer ejecta) while differing in NUV flux (set by IGE in the inner ejecta). *The NUV flux ratio between optical twins encodes precisely the information needed to break the luminosity degeneracy.*

2 Proposed Research Program

This program connects theoretical spectral modeling, observational NUV surveys, cosmological distance measurement, and—newly—a first-principles, quantum-accelerated explosion-physics module that grounds the empirical subclass calibration in the underlying nuclear and combustion physics.

Pillar I | Spectral Analysis with Lumina-SN / Lumina-ML

Pillar II | NUV Observations with S-Drift

Pillar III | Refined Absolute Luminosity & the Hubble Law

Pillar IV | Quantum-Accelerated Explosion Physics (DDT Module)

2.1 Pillar I: GPU-Accelerated Spectral Analysis

Two codes I have developed anchor this pillar: **Lumina-SN**, a C/CUDA Monte-Carlo radiative transfer code, and **Lumina-ML**, a neural spectral-emulation and Bayesian-inference framework.

Lumina-SN (the forward model) solves 1D Monte-Carlo radiative transfer in homologously expanding ejecta under the Sobolev approximation: photon-packet emission from a sharp photosphere at v_{ph} , propagation through radial shells with a broken power-law density profile, and interactions via Thomson scattering and bound-bound (Sobolev) line opacity; iterative convergence of the radiation field (W, T_{rad}) with an LTE plasma state over 5–20 cycles; and a three-zone (Core/Wall/Outer) ejecta structure with tunable density exponents and composition (Si, Fe, O, C, Ca, S, Mg, Ni). The C/CUDA implementation achieves $\sim 10\times$ speedup over TARDIS on NVIDIA GPUs, enabling the 10^4 – 10^5 forward models required for inference.

Lumina-ML (neural emulation & inference). A single Lumina-SN evaluation costs ~ 1 s; to enable full Bayesian inference over a 15-D parameter space, Lumina-ML builds a neural emulator: (1) 5×10^4 spectra on a Latin-hypercube design; (2) PCA compression of each spectrum (~ 1100 bins $\rightarrow \sim 60$ components, $>99.9\%$ variance); (3) an MLP ($15 \rightarrow 60$, ~ 500 K params, SiLU) mapping parameters to PCA coefficients in $\sim 10 \mu\text{s}$ —five orders of magnitude faster than the simulation; and (4) inference via MCMC (`emcee`), nested sampling (`dynesty`), and simulation-based inference (SBI/SNPE).

Application to the Twin Problem

Given an observed SN Ia spectrum, Lumina-ML returns the full posterior over 15 physical parameters. For a twin pair we quantify exactly which parameters differ—total ^{56}Ni mass, iron-group distribution, or progenitor metallicity—providing the physical diagnosis that underpins an improved luminosity calibration.

2.2 Pillar II: NUV Observations with S-Drift

The NUV (2000–3500 Å) is inaccessible from the ground, requiring space-based observation. The **S-Drift** UV space telescope provides an ideal platform. The observational program will: obtain NUV photometry (~ 2500 Å) and low-resolution spectroscopy for a volume-limited sample ($z < 0.05$) from ZTF / Rubin-LSST; identify optical-twin pairs and measure their NUV flux ratios; construct the first statistically significant NUV–optical color–luminosity relation; and cross-correlate NUV properties with host-galaxy mass, metallicity, and star-formation rate to test the progenitor-metallicity hypothesis.

2.3 Pillar III: Refined Absolute Luminosity and the Hubble Law

The goal is a subclass-aware calibration that replaces the single Phillips relation with a multi-parameter standardization incorporating NUV information:

$$M_B = M_0 + \alpha (\Delta m_{15} - 1.1) + \beta (B - V)_{\text{max}} + \gamma \Delta \text{NUV} + \delta \theta_{\text{Fe}}, \quad (1)$$

where ΔNUV is the NUV color excess (from S-Drift) and θ_{Fe} is the iron-group parameter from Lumina-ML fitting. Adding these two parameters is expected to reduce the Hubble residual scatter from $\sigma \approx 0.12$ mag to $\sigma \lesssim 0.08$ mag—a $\sim 30\%$ improvement in per-object distance precision—directly tightening H_0 and $w(z)$.

3 Pillar IV: Quantum-Accelerated Explosion Physics

The empirical subclass calibration (Pillars I–III) is most powerful when grounded in the *physics that creates the subclasses*. The deflagration-to-detonation transition (DDT) in the Chandrasekhar-mass channel sets the transition density ρ_{DDT} , the synthesized ^{56}Ni mass, and the radial IGE distribution—precisely the quantities that drive the NUV divergence of optical twins. Two pieces of the DDT problem are computationally dominant and map naturally onto the *two distinct quantum architectures* now available to this program: **IonQ** (trapped-ion gate model) and **D-Wave** (quantum annealer).

Framing. Pillar IV is deliberately scoped as a *NISQ-era methods-and-feasibility thrust*, not a claim of near-term quantum advantage or near-term astrophysical constraints on H_0 . Each quantum workflow is paired with a strong classical baseline and judged by reusable deliverables (QUBO formulations, benchmark datasets, sensitivity studies) and by its impact on Lumina-SN calibration products. The

same surrogate / QUBO pipelines remain scientifically useful as classical tools should the quantum hardware underperform—an explicit fallback that de-risks the pillar.

3.1 Two Architectures, Cleanly Separated

IonQ – gate-based, trapped-ion

- ✓ Arbitrary quantum circuits (VQE, QAOA, QML, Trotter)
- ✓ All-to-all connectivity, high fidelity ($\sim 99.x\%$ 2-qubit)
- ✗ Limited width: ~ 25 – 36 algorithmic qubits (Aria/Forte)
- $\Rightarrow$ “small but precise” quantum chemistry / nuclear / ML

D-Wave – quantum annealer

- ✓ 5000+ qubits; large combinatorial optimization & sampling
- ✓ Native QUBO / Ising; reverse-anneal sampling
- ✗ QUBO form only (no gate circuits $\Rightarrow$ no VQE/QPE/HHL)
- $\Rightarrow$ “large but shape-restricted” optimization / inverse / classification

Guiding Principle

DDT *microphysics* (nuclear reaction rates) $\rightarrow$ **IonQ**. DDT *combinatorial & inverse problems* (ignition placement, parameter estimation) $\rightarrow$ **D-Wave**. Do not mix the roles.

3.2 D-Wave: The Combinatorial & Inverse Half of the DDT

IV.1 Multi-Spot Ignition-Kernel Placement via a Quadratic Surrogate

In DDT-class models the initial deflagration ignition geometry—how many kernels to seed, and where in the white-dwarf core—governs the ^{56}Ni mass and its radial distribution, and hence the light curve and NUV color. This is a *combinatorial placement* problem over binary occupation variables $x_i \in \{0, 1\}$ (kernel at a physics-filtered candidate site i). The full radiation-hydro response $L_\lambda(\mathbf{x})$ is nonlinear in $\mathbf{x}$ and is *not* directly a QUBO; we therefore fit a *quadratic response surrogate* over the candidate sites,

$$L_\lambda^{\text{model}}(\mathbf{x}) \approx L_\lambda^0 + \sum_i a_{\lambda i} x_i + \sum_{i < j} b_{\lambda ij} x_i x_j, \quad (2)$$

with coefficients learned from a designed ensemble of classical 1D/2D explosion models. The resulting objective $\sum_\lambda (L_\lambda^{\text{obs}} - L_\lambda^{\text{model}})^2 + \lambda_1 (\sum_i x_i - N_k)^2 + \lambda_2 \sum_{\langle ij \rangle} J_{ij} x_i x_j$ (count and minimum-separation penalties) is then an explicit QUBO. The D-Wave annealer *samples low-energy configurations* of this quadratic approximation inside an *outer classical loop* that updates the surrogate and validates candidate placements with full Lumina-SN evaluations.

Most actionable near-term use case

Surrogate-based ignition placement is the most actionable near-term quantum-hardware task in this program. Because quantum annealers have not demonstrated a general optimization advantage for this problem class, *all* D-Wave results will be benchmarked on identical surrogate objectives against strong classical methods—simulated annealing, parallel tempering, tabu search, Bayesian optimization, and (where tractable) exact / mixed-integer quadratic optimization—with success measured by solution quality, time-to-target, and solution diversity.

IV.2 DDT-Parameter Inverse Estimation

Continuous DDT parameters—the transition density ρ_{DDT} , the Karlovitz-number threshold, etc.—are recovered by binary expansion into QUBO variables and minimizing the model–observation (spectrum / light-curve) mismatch. Reverse annealing is used as an *approximate local sampler / proposal generator* for this multimodal inverse problem—*not* assumed to be a calibrated Bayesian posterior; posterior claims are made only after

validation against classical MCMC / nested-sampling baselines on test problems with known targets, before combining with ZTF/LSST ensemble statistics.

IV.3 QBoost — a DDT-Precursor Classifier

Using existing 3D explosion simulations as labelled training data, a classifier for “where DDT ignites” is learned with D-Wave **QBoost** (sparse weak-learner ensemble selection cast as a QUBO), benchmarked against logistic regression, random forests, and gradient boosting (XGBoost) under class imbalance (AUROC/AUPRC, false-positive rate at fixed recall). The QUBO’s sparse learner selection is valued for *interpretability* of the precursor indicators, not assumed accuracy advantage.

3.3 IonQ: The Nuclear-Microphysics Half of the DDT

IV.4 VQE — Restricted Nuclear-Structure Benchmark for the DDT Chain

The DDT density ρ_{DDT} depends on the induction-time gradient (Zel’dovich gradient mechanism), and τ_{ind} is strongly temperature- and rate-sensitive to the $^{12}\text{C}+^{12}\text{C}$ reaction. We scope the IonQ task *honestly*: a VQE on 25–36 noisy qubits cannot produce a predictive $^{12}\text{C}+^{12}\text{C}$ astrophysical S -factor (a sub-barrier fusion problem involving tunneling, resonances, compound- ^{24}Mg states, channel couplings, and partial widths). Instead, VQE will target a *restricted active-space shell-model Hamiltonian* for selected ^{24}Mg levels (Jordan–Wigner mapping; IonQ’s all-to-all connectivity suits the dense nuclear term structure), benchmarked against *exact diagonalization* in the same truncated space. The physically meaningful chain is therefore:

VQE restricted spectrum $\rightarrow$ benchmarked level-energy sensitivity $\rightarrow$ perturbation of a conventional reaction-rate model

not a standalone quantum prediction of the rate.

Deliverable

A demonstrated workflow propagating controlled nuclear-Hamiltonian perturbations into τ_{ind} *sensitivity* (“if a level/rate shifts by $X\%$, the DDT criterion moves by . . .”), anchored to established classical reaction-rate calculations—a sensitivity-informed correction, not a production $\tau_{\text{ind}}(\rho, T, X)$ table. This preserves the link to the Zel’dovich mechanism while remaining honest about NISQ and model-space limitations.

IV.5 Variational Quantum Classifier (QML)

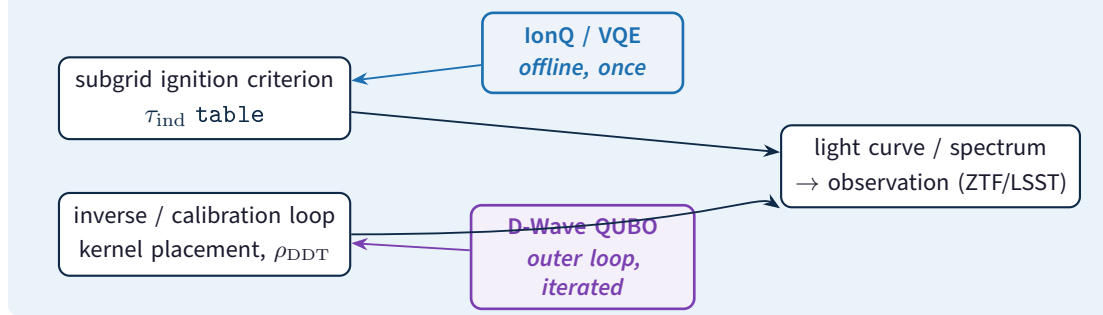
A variational quantum classifier (VQC) learns DDT-precursor patterns as a *diagnostic benchmark* against the D-Wave QBoost classifier (IV.3) and classical ML—a publishable annealer-vs-gate study. On small noisy devices a VQC is not expected to beat classical ML on tabular data; its value is the controlled cross-architecture comparison, not a flagship accuracy claim.

IV.6 QAOA — Small-Scale Ignition Placement (annealer benchmark)

The same ignition-placement QUBO (IV.1) solved on IonQ via **QAOA** provides a direct *annealer-vs-gate* performance comparison—a methodology contribution in its own right.

3.4 Hybrid Integration with the Classical SN Ia Code

Classical SN Ia code (under development): hydro + flame-tracking + nuclear network + radiative transfer



- **lonQ output = offline lookup table:** per-timestep quantum calls are infeasible (latency, queueing); reaction rates / τ_{ind} are precomputed and tabulated into the code.
- **D-Wave output = external calibration loop:** the explosion simulation is wrapped in a surrogate, and the annealer searches parameters / ignition placement around it.

3.5 Minimum-Viable Starting Projects

Project A — D-Wave (recommended first; faster results)

Optimize multi-spot ignition placement (quadratic surrogate) to match an observed light curve.

1. Build a placement $\rightarrow$ light-curve quadratic surrogate from 1D/2D models.
2. Encode the surrogate mismatch as a QUBO.
3. Submit via the Ocean SDK (`dwave-system`, `dimod`).
4. Benchmark vs. simulated annealing / parallel tempering.

Project B — lonQ

VQE benchmark of a restricted ^{24}Mg active-space spectrum.

1. Select a small active-space shell-model Hamiltonian.
2. Build the VQE (Qiskit-Nature / PennyLane).
3. Submit on the lonQ backend (Qiskit-IonQ / PennyLane-IonQ).
4. Validate vs. exact diagonalization; simulator $\rightarrow$ hardware.

The Ocean SDK has a gentle learning curve, with first results expected within a few weeks.

3.6 Evaluation, Baselines, and Resource Accounting

Each quantum task carries a pre-registered classical baseline and quantitative success criteria, so that “does quantum help here?” is answered, not assumed:

- **Baselines:** simulated annealing, parallel tempering, tabu search, Bayesian optimization, and MILP/MIQP (D-Wave / QAOA); exact diagonalization and classical shell-model codes (VQE); logistic regression, random forests, XGBoost (classifiers).
- **Success metrics:** best objective value and time-to-target on identical surrogate objectives; solution diversity and robustness to surrogate noise; reproducibility across anneals; VQE energy error vs. exact diagonalization; AUROC/AUPRC under imbalance; and—ultimately—the change in Lumina-SN spectral / light-curve residuals.
- **Resource accounting:** number of binary variables, QUBO density, coefficient precision, D-Wave minor-embedding overhead and chain-strength strategy (logical vs. physical qubits); for lonQ, circuit depth, shot budgets, and measurement / symmetry-based error mitigation.
- **Surrogate validity:** $L_{\lambda}^{\text{model}}(\mathbf{x})$ is trained, cross-validated, and locally re-quadrated within the outer loop, with surrogate-to-QUBO uncertainty propagated.

3.7 Honest Caveats

- **D-Wave cannot run VQE/HHL.** Linear-system and reaction-network solves belong to IonQ (or future FTQC). Keep the architectural roles separate.
- **IonQ qubit limits** strongly constrain the nuclear model space—initial goals target *demonstrated quantum feasibility*, not yet quantitative accuracy.
- **Quantum CFD** (direct turbulent-flame simulation) is beyond both machines—an FTQC-era problem, deferred.
- **Latency:** cloud-queue waits make in-line simulation calls impractical on both platforms $\Rightarrow$ the offline-table / external-loop pattern is mandatory.

4 Methodology and Timeline

Phase	Activities
Phase 1 Year 1	Generate $\sim 10^5$ Lumina-SN synthetic spectra spanning the SN Ia parameter space (emphasis on IGE abundance/distribution); train and validate the Lumina-ML emulator ($< 1\%$ RMS, 2000–10000 Å); apply Bayesian inference to archival spectra (CfA/CSP/Berkeley). Quantum kickoff: Project A (D-Wave ignition-placement QUBO) and Project B (IonQ VQE proof-of-concept) on simulators.
Phase 2 Year 1–2	Automated optical-twin identification (3800–7200 Å cross-correlation, ~ 3000 spectra); per-twin 15-parameter posteriors and identification of the luminosity-discrepancy driver; NUV predictions for S-Drift. Quantum: D-Wave hardware runs + classical-SA benchmark; IonQ hardware validation; τ_{ind} table v1.
Phase 3 Year 2–3	S-Drift NUV photometry/spectroscopy of new SNe Ia and twin candidates; validate Lumina-ML predictions; build the subclass-aware calibration (Eq. 1) and quantify the scatter improvement. Quantum: couple τ_{ind} corrections + ignition placement into the explosion module; annealer-vs-gate (QAOA) comparison study.
Phase 4 Year 3	Apply the refined calibration to Pantheon+ / Union; derive updated $H_0, w(z)$; assess Hubble-tension relief; prepare for Rubin/LSST and Roman.

5 Technical Readiness and Resources

- **Lumina-SN** is operational, validated against TARDIS to sub-1% (W error 0.44%, T_{rad} error 0.36%), and demonstrated on SN 2011fe (best-fit RMS = 0.0894). The C/CUDA build achieves $\sim 10\times$ GPU speedup (200 K packets: 728 ms GPU vs. 7.2 s CPU).
- **Lumina-ML** provides a complete emulation + inference pipeline (MCMC, nested sampling, SBI); the emulator evaluates spectra in $\sim 10 \mu\text{s}$.
- **Computing:** KIAS CAC dedicated GPU clusters (NVIDIA A100/H100); additional KISTI Nurion/Neuron allocations via the Grand Challenge Program.
- **Quantum access:** IonQ (gate-model, via cloud providers) and D-Wave (Leap / Ocean SDK) for Pillar IV; both used in the offline-table / external-loop pattern above.
- **Collaborations:** KASI supernova group, the FOSTER multi-wavelength group, and the SDSS Korean Scientist Group for spectral databases.

6 Expected Outcomes and Impact

1. **Physical classification of SN Ia subclasses:** the first systematic mapping between spectroscopically inferred parameters (IGE mass/distribution, progenitor metallicity) and the optical-twin / NUV-divergence

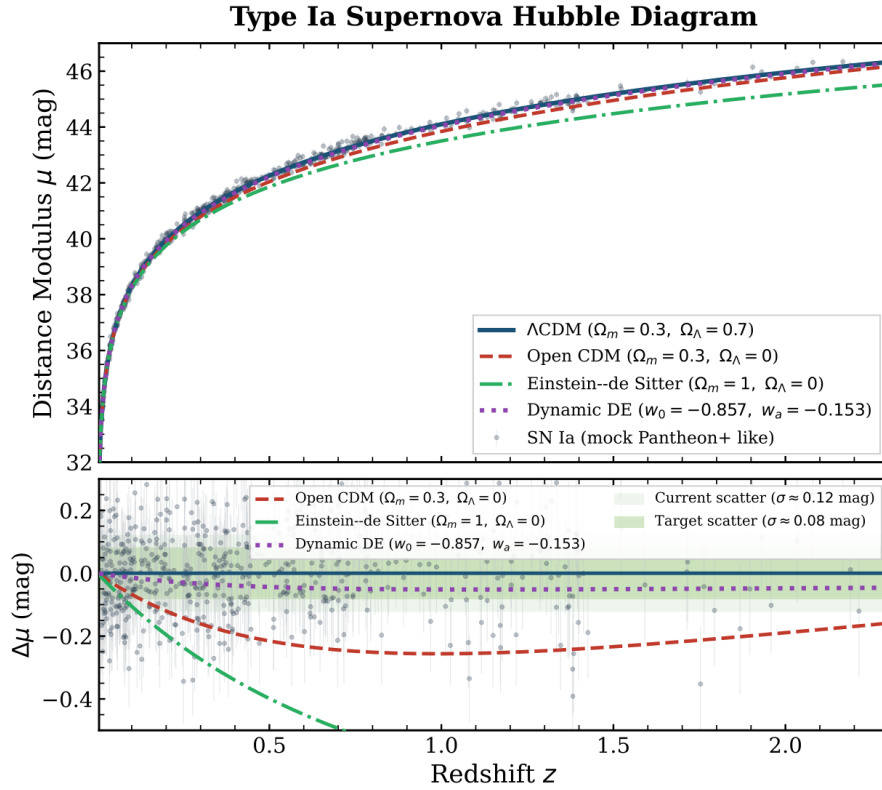


Figure 2: Type Ia Supernova Hubble Diagram. *Upper panel:* Distance modulus μ vs. redshift z for SN Ia observations (mock data representative of the Pantheon+ compilation) compared with theoretical predictions from four cosmological models: the standard Λ CDM ($\Omega_m = 0.3, \Omega_\Lambda = 0.7$; solid blue), an open CDM universe ($\Omega_m = 0.3, \Omega_\Lambda = 0$; dashed red), the Einstein-de Sitter model ($\Omega_m = 1, \Omega_\Lambda = 0$; dot-dashed green), and a dynamic dark energy model with CPL parameterization ($w_0 = -0.857, w_a = -0.153$; dotted purple). *Lower panel:* Residuals $\Delta\mu$ relative to Λ CDM. The shaded bands indicate the current per-SN scatter ($\sigma \approx 0.12$ mag; light green) and our target scatter after NUV-augmented calibration ($\sigma \approx 0.08$ mag; darker green).

Figure 2: **Type Ia Supernova Hubble diagram.** *Upper:* distance modulus vs. redshift for mock Pantheon+-like data against four cosmologies (Λ CDM; open CDM; Einstein-de Sitter; CPL dynamic dark energy $w_0 = -0.857, w_a = -0.153$). *Lower:* residuals $\Delta\mu$ relative to Λ CDM; bands show the current per-SN scatter ($\sigma \approx 0.12$ mag) and the target after NUV-augmented calibration ($\sigma \approx 0.08$). Distinguishing Λ CDM from dynamic dark energy requires sub-0.1-mag precision—motivating the improved standardization proposed here.

phenomenon.

2. **NUV-augmented luminosity calibration:** a new standardization (Eq. 1) reducing Hubble scatter by $\sim 30\%$ ($\sigma \approx 0.12 \rightarrow \lesssim 0.08$ mag).
3. **Improved H_0 :** tighter local distance-ladder constraints bearing directly on the Hubble tension.
4. **First-principles explosion grounding:** a quantum-accelerated DDT module linking nuclear microphysics (τ_{ind}) and ignition combinatorics to the observable subclasses—plus standalone methodology results (annealer-vs-gate; QBoost-vs-VQC).
5. **Public code release:** Lumina-SN, Lumina-ML, the trained emulator, the template library, and the quantum-module scaffolding.
6. **S-Drift synergy:** a key science case for the S-Drift UV mission.

7 Cosmological Applications: H_0 Tension and Dynamic Dark Energy

7.1 The Hubble Tension

The $\sim 5\sigma$ discrepancy between the local H_0 (73.0 ± 1.0 km/s/Mpc, SH0ES; Riess et al. 2022) and the CMB-inferred value (67.4 ± 0.5 km/s/Mpc, Planck 2020) remains unresolved. A significant share of the local error budget originates in the SN Ia standardization step: if unrecognized subclasses with different luminosities are mixed into the calibrator sample, H_0 may be biased. Our NUV-augmented calibration will quantify whether the SH0ES calibrator sample contains a NUV-red/NUV-blue mixture, and whether correcting it shifts H_0 —providing a cleaner, subclass-homogeneous distance ladder and reducing H_0 systematics below the current $\sim 1.4\%$ level.

7.2 Dynamic Dark Energy and the Equation of State $w(z)$

Unrecognized astrophysical systematics in the SN Ia Hubble diagram can mimic or mask dynamic dark energy. Our subclass-aware calibration removes this systematic, enabling a clean test of $w(z) \neq -1$. Combining improved SN Ia distances with DESI/SDSS BAO and Planck/ACT CMB constraints, we will construct a joint analysis simultaneously constraining H_0 , Ω_m , and the CPL parameters (w_0, w_a)—the most stringent test to date of whether dark energy is a cosmological constant or an evolving component.

In summary, this proposal bridges computational spectral modeling (Lumina-SN/ML), space-based NUV observation (S-Drift), precision cosmology, and a first-principles, quantum-accelerated explosion-physics module (IonQ + D-Wave). By revealing and quantifying the physical substructure hidden within the Phillips relation—and grounding it in the nuclear and combustion physics that creates it—we aim to improve SN Ia distance precision by $\sim 30\%$, directly contributing to the resolution of the H_0 tension and to constraining the time evolution of dark energy.